Process Flow

PROCESS NAME:	BSI.IT.23 Systems Development Life Cycle (SDLC)

DETAILED PROCESS DESCRIPTION

The System Development Life Cycle (SDLC) is a process around planning, developing, testing, and deploying an information system in BSI's environment. It is utilized for system and software development projects for internal systems and software.

CURRENT STATUS	Draft
DATE APPROVED:	
APPROVED BY:	

VERSION HISTORY	VERSION	STATUS	COMPLETION DATE
	1.0	Draft	11/8/25

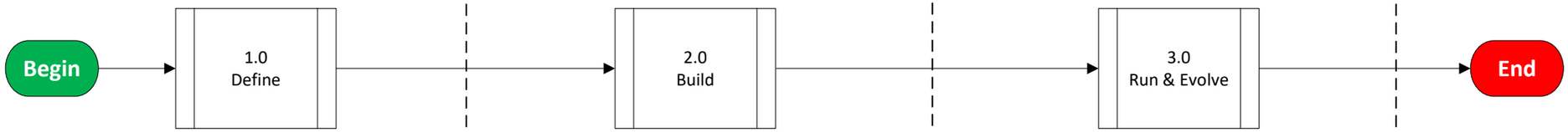
REVIEW AND CHALLENGE

NAME	ROLE
Elizabeth Noda	BUSINESS CONTROL OFFICER
Jose Luis Martinez	PROCESS OWNER
	SUBJECT MATTER EXPERT

Systems Development Life Cycle (SDLC)

High Level

Overview



Key Processes

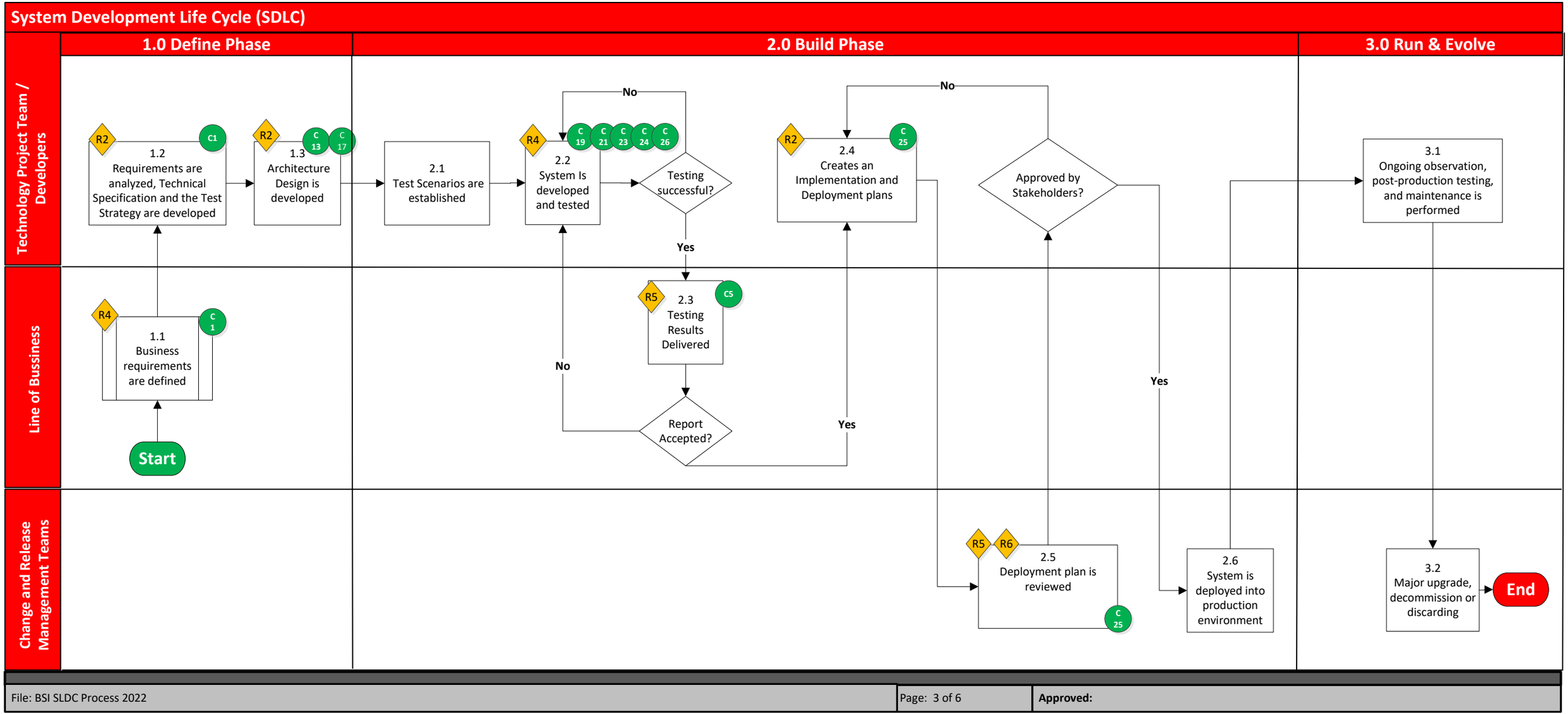
1.0 Define	2.0 Build	3.0 Run & Evolve
• Business requirements are defined • Requirements are analyzed, Technical Specification and the Test Strategy are developed • Architecture Design/ISAR is developed	• Test Scenarios are established • System Is developed and tested • Implementation and deployment plans are defined • Deployment plan is reviewed	• System deployed into production environment • Ongoing observation, post-production testing, and maintenance is performed • Major upgrade, decommission or discarding

People

1.0 Define	2.0 Build	3.0 Run & Evolve
• Technology Project Team/ Developers • Line of Business	• Technology Project Team/ Developers • Line of Business • Change & Release Management Teams	• Technology Project Team/ Developers • Line of Business • Change & Release Management Teams

Timing/SLA/OLA

1.0 Define	2.0 Build	3.0 Run & Evolve
• In every operation	• In every operation	• In every operation


System Development Life Cycle (SDLC)
1.0 Define Phase
2.0 Build Phase
3.0 Run & Evolve
Technology Project Team / Developers
Line of Bussiness
Change and Release Management Teams
R2
C1
1.2
Requirements are analyzed, Technical Specification and the Test Strategy are developed
R2
C 13
C 17
1.3
Architecture Design is developed
2.1
Test Scenarios are established
R4
C 19
C 21
C 23
C 24
C 26
2.2
System Is developed and tested
No
Testing successful?
Yes
R5
C5
2.3
Testing Results Delivered
Report Accepted?
No
Yes
R2
C 25
2.4
Creates an Implementation and Deployment plans
No
Approved by Stakeholders?
Yes
3.1
Ongoing observation, post-production testing, and maintenance is performed
R4
C 1
1.1
Business requirements are defined
Start
R5
R6
2.5
Deployment plan is reviewed
C 25
2.6
System is deployed into production environment
3.2
Major upgrade, decommission or discarding
End

Controls

- [BSI.IT.23.C1 Business Requirement] As part of developing a new internal system, customizing vendor products or upgrading existing systems, functional and non-functional requirements must be defined and approved by the business sponsor prior to entering the Build phase of the SDLC process.
- [BSI.IT.23.C13 Design of System] As part of developing a new internal system, customizing vendor products, upgrading existing systems, or onboarding a new technology, the Project Team submits the Architecture Design artifacts for review and approval by Enterprise Architecture and submits the Data (logical, physical) model, Data artifacts mandated by Chief Data Officer (CDO) if the project had data impact for review and approval by the Data Governance team(CDO).
- [BSI.IT.23.C17 Solution Intent Doc] As part of developing a new internal system, customizing vendor products or upgrading existing systems, if Book of Work criteria is met, the Solution Intent document must be reviewed and approved by the Architecture Review Board (ARB) prior to entering the development phase of the SDLC process.
- [BSI.IT.23.C21 SIT] Software Development team performs system integration testing (SIT) for new developments with integrations, which is reviewed and approved by the Technical Project Team prior to production implementation. Issues identified in testing are addressed prior to production implementation.
- [BSI.IT.23.C23 Pen Test] For newly developed applications, Software developers address or mitigate issues identified by Information Security through penetration tests prior to production implementation. Information Security must approve the testing results prior to production implementation.
- [BSI.IT.23.C24 UAT] For internally developed applications, Application owners perform and approve User Acceptance Testing (UAT) prior to production implementation.
- [BSI.IT.23.C25 Implement & Deployment] Software developers document the Implementation Plan for deployment of code into production, which are reviewed and approved by relevant stakeholders prior to production implementation.
- [BSI.IT.23.C26 App Sec Test] Software Developers mitigate issues identified in Dynamic Code analysis scans as required by Information Security prior to implementation to production.
- [BSI.IT.23.C4 Segregated Environments] Software Development and QA teams develop and test code in non-production environments. Exceptions must be approved by Information Security.

NON-KEY

- [BSI.IT.23.C19 Unit Testing] Software developers perform unit testing of code to identify and correct issues prior to entering the run phase of the SDLC process.

Systems Development Life Cycle (SDLC)

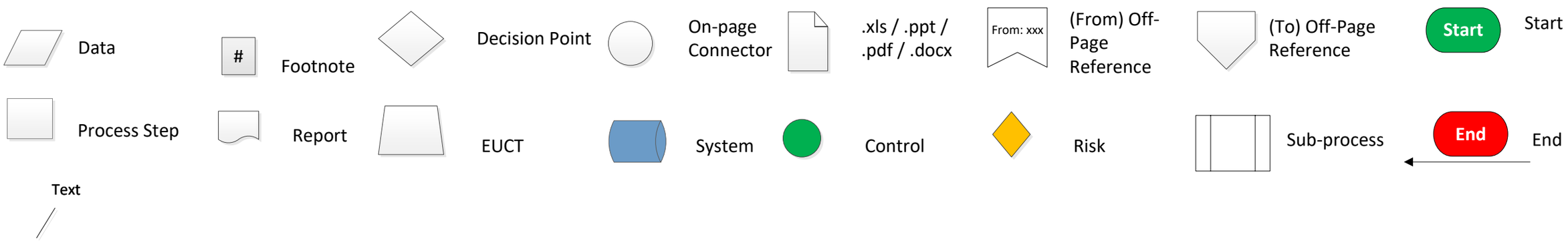
Risks

Regulations	Risks	Systems	Vendors
• GLBA	• R2 - [BSI.IT.23.R2 Design Architecture] Instability in the production architecture caused by lack of review and flawed design implementation may result in production issues and business disruption. C17 • R3 - [BSI.IT.23.R3 Inadequately Configured Sys] Inappropriate data disclosures caused by the production version into the IT environment resulting in production issues, business disruptions and regulatory non-compliance. C4 • R4 - [BSI.IT.23.R4 Lack of Design Phase] Insufficient identification and documentation of business and functional requirements for new or updated system designs, due to poorly established design phase process requirements, may result in business production issues and regulatory non-compliance. C1, C13 • R5 - [BSI.IT.23.R5 Lack of Testing] Insufficient identification of software security, audit process and control requirements, caused by a lack of complete testing performed consistently, may result in production issues, business disruption and regulatory noncompliance. C19, C21, C23, C24, C26 • R6 - [BSI.IT.23.R6 Improper Coding Practices] Software is vulnerable to malicious attacks and service disruptions due to improper coding practices and planning which may result in exploitable vulnerabilities and configurations, business disruptions, financial loss, and regulatory noncompliance.C25	• xx	• xx

Systems Development Life Cycle (SDLC)

Legend and Notes

Legend



Notes and Comments